How Topological Surface States Affect Thermoelectric Properties, Measured in $(Bi_{1-x}In_x)_2Se_3$ Thin Films Using Arduino and LabVIEW

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Abstract

Thermoelectric materials can convert thermal energy into electrical energy, making them great candidates for harvesting waste heat, e.g. from data centers. Currently, many of the best thermoelectrics happen to be topological insulators, which are a new type of insulator with topological surface states that make only the surface conducting. They are being actively studied for a wide range of applications, including topological quantum computing and spintronics. However, whether topological surface states enhance or degrade thermoelectric properties remains unknown. To address this, an efficient and economical thermoelectric measurement setup was developed using Arduino and LabVIEW. This setup could measure multiple thermoelectric properties like the Seebeck coefficient S, the conductivity σ, and the thermoelectric power factor $S^2\sigma$ simultaneously over a range of temperatures. With this setup, the thermoelectric properties in topologically tunable $(Bi_{1-x}In_x)_2Se_3$ thin films were measured as topological surface states were gradually removed by increasing the indium concentration (x). It was discovered that the magnitude of the Seebeck coefficient increased as the surface states were removed, while the conductivity and power factor (and therefore the thermoelectric performance) decreased. This suggests that topological surface states enhance thermoelectric properties; in other words, pure Bi_2Se_3 exhibits the best thermoelectric performance. Hence, this study reveals the critical role of topological surface states in thermoelectricity and sheds light on the path toward better thermoelectric materials for recovering waste heat.